SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1 ✓
Secure Employee Record Management System
50 marks

Q2 ✓
Bank Transaction File and Database System
50 marks

2/2 answered

Q1 Secure Employee Record Management System 50 marks

Develop a Java application for an organization to securely manage employee records using hashing and an appropriate hash function for employee identification. Store employee details in a file using byte/character streams and implement serialization for storing employee objects. Provide options to add, search, update, and delete employee records. The system should handle duplicate IDs and invalid file operations appropriately.

Your Code

```java
import java.util.*;
import java.security.MessageDigest;

public class Employee {
    String id;
    String name;
    String department;
    double salary;

    Employee(String id, String name, String department, double salary)
        this.id = id;
        this.name = name;
        this.department = department;
        this.salary = salary;
    }

    static String hash(String id) throws Exception {
        MessageDigest md = MessageDigest.getInstance("SHA-256");
        byte[] bytes = md.digest(id.getBytes("UTF-8"));

        StringBuilder result = new StringBuilder();

        for (byte b : bytes) {
            result.append(String.format("%02x", b));
        }
```

Input

```
101
Rahul
IT
50000
```

Output

```
Employee added successfully
ID: 101
Name: Rahul
Department: IT
Salary: 50000.0
```

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QUESTIONS

- **Q1** Secure Employee Record Management System — 50 marks
- **Q2** Bank Transaction File and Database System — 50 marks

2/2 answered

Q2 Bank Transaction File and Database System

50 marks

Design a banking application that maintains customer and transaction information using Java I/O and JDBC. Store transaction logs using stream I/O and serialize customer objects for backup. Establish JDBC connectivity with a relational database and implement INSERT, UPDATE, DELETE, and SELECT operations using appropriate Statement types. The application must validate transactions and handle database and I/O exceptions.

Your Code

```java
import java.io.*;
import java.util.*;

public class BankTransaction implements Serializable {

    static class Customer implements Serializable {
        int id;
        String name;
        double balance;

        Customer(int id, String name, double balance) {
            this.id = id;
            this.name = name;
            this.balance = balance;
        }
    }

    static ArrayList<Customer> customers = new ArrayList<>();

    static void insert(int id, String name, double balance) {
        for (Customer c : customers) {
            if (c.id == id) {
                System.out.println("Duplicate customer ID");
```

Input

```
1
101
Rahul
50000
```

Output

```
Customer inserted
```